

Author Disclosure: S.E. Finkelstein: G. Other; Co-inventor of technology licensed to company. A. Trotti: None. G.D. Letson: None. M. Russell: None. T.F. DeLaney: None. K.M. Alektiar: None. J.M. Michalski: None. D. Wang: None. T. Kadir: A. Employment; Mirada Medical. C.W. Stevens: None.

2993 Difference in Rates of Necrosis in Patients with Primary Soft Tissue Sarcoma of the Extremity and Trunk Treated with Neoadjuvant Radiotherapy or Chemoradiotherapy

M. Bedi, E. V. Zambrano, D. M. King, D. A. Hackbarth, J. Charlson, R. Whitfield, C. Bergom, D. Wang

Medical College of Wisconsin, Milwaukee, WI

Purpose/Objective(s): Neoadjuvant radiotherapy (NRT) or chemoradiotherapy (NCRT) followed by wide local resection yields excellent local control rates in the treatment of soft tissue sarcomas (STS). Extensive necrosis on the resected surgical specimen following neoadjuvant treatment is often viewed as objective treatment response. We aimed to determine whether the rates of histopathologic necrosis in STS are different following NRT vs NCRT.

Materials/Methods: From May 1999 to July 2010, 112 patients (pts) with primary localized STS of the extremity and trunk received either NRT or NCRT followed by surgical wide local excision. Histology slides and pathology reports from 103 pts were available for central review. Tumor characteristics examined and centrally reviewed included histology, grade and percent (%) of necrosis defined using the Willet criteria.

Results: Out of 103 pts, 59 (57%) received NRT and 44 (43%) received NCRT. Median radiation dose was 50.4 Gy. Chemotherapy was an adriamycin-ifosfamide based regimen, with 38 pts receiving chemotherapy followed by RT and 6 pts receiving 3 cycles of chemotherapy interdigitated with RT. Of the 59 pts treated with NRT, 60% had high grade, 10% had intermediate grade, and 30% had low grade disease. Of the 44 pts treated with NCRT, 95% had high grade, 5% had intermediate grade, and 0% had low grade disease. Median tumor size for NRT vs NCRT group was 7.6 cm and 9.6 cm, respectively. Overall, there was a significant difference in median % necrosis between the NRT and NCRT group (50% vs 80%, $p = 0.03$). Three pts (5.26%) in the NRT and 8 pts (17%) in the NCRT group had 100% necrosis on final pathology ($p = 0.05$). However, there were no differences in the common histological subtypes between NRT and NCRT group. Median % necrosis observed in the common histology subtypes in these two groups are noted in Table 1.

Conclusions: Overall % necrosis on final pathology was significantly higher in pts with primary STS of the extremity and trunk treated with NCRT compared with NRT. In addition, there is a suggestion that complete necrosis is also higher in the NCRT group. This increase might be related to chemotherapy effect although other factors such as high grade and tumor size (greater in the NCRT group) remain to be defined.

Histology	No. of Patients	Median % necrosis: NRT	Median % Necrosis: NCRT
MFH	NRT: 16 NCRT: 15	75	90
Liposarcoma	NRT: 15 NCRT: 9	28	93
Myxofibrosarcoma	NRT: 10 NCRT: 6	80	97
Leiomyosarcoma	NRT: 6 NCRT: 2	68	60
Synovial Sarcoma	NRT: 1 NCRT: 6	1	10
Extraskelatal Myxoid Chondrosarcoma	NRT: 3 NCRT: 1	1	40
MPNST	NRT: 3 NCRT: 1	40	10
Fibrosarcoma	NRT: 1 NCRT: 1	0	80
Spindle Cell Sarcoma	NRT: 1 NCRT: 1	50	20

Author Disclosure: M. Bedi: None. E.V. Zambrano: None. D.M. King: None. D.A. Hackbarth: None. J. Charlson: None. R. Whitfield: None. C. Bergom: None. D. Wang: None.

2994 Prognostic Factors in Patients with Choroidal Melanoma Treated with Iodine 125 Brachytherapy

S. Garg, S. Reich, B. Gielda, B. Dilman, D. Bernard, J. Cohen, A. Herskovic

Rush University Medical Center, Chicago, IL

Purpose/Objective(s): Temporary Iodine 125 (I-125) plaque brachytherapy is an accepted treatment for patients with intermediate risk choroidal melanoma. In order to identify prognostic features in this disease, we reviewed outcomes of treated patients with respect to presenting ocular exam findings, tumor characteristics, total dose and dose rate received. Outcomes reviewed included development of metastases and salvage enucleation.

Materials/Methods: A retrospective chart review was performed of consecutive patients with choroidal melanoma treated with I-125 between 1985 and 2008 at a single tertiary care center. Data collected included: patient age, tumor size, presence of retinal detachment (RD) at diagnosis, delivered dose and dose rate. Univariate and multivariate analyses were performed to assess the ability of factors to predict patient outcome, defined by development of metastases or requirement for enucleation.

Results: 79 patients with choroidal melanoma were treated and followed for at least 6 months, with a mean follow-up of 39.9 months. Mean age at presentation was 64.8 years. The mean total dose delivered to the tumor apex was 88.6Gy and the mean dose rate was 72.2 cGy/hr. Ultimately, 8 patients (10.1%) underwent enucleation for local failure and 8 (10.1%) developed metastatic disease. One of the patients with metastatic disease was also enucleated. Total dose and dose rate were not associated with progression to enucleation or metastases on univariate or multivariate analysis. Data regarding patterns of failure with respect to location within the irradiated volume are under analysis. There was a significant difference between the median largest tumor dimension for both patients who were enucleated (15.5mm, $p = 0.0003$) and who developed metastatic disease (14.8mm, $p = 0.01$) versus those who did not (11.6mm). RD was identified in 19 (24%) patients at diagnosis. Of this group, 6 patients developed metastases, including 1 patient who was also enucleated. On univariate analysis, patients with RD on presentation were 13 times more likely than those without RD to progress to metastatic disease (OR = 13.4). Upon multivariate analysis, the presence of RD was still a statistically significant predictor of the development of metastatic disease ($p = 0.03$).